Hold Out Test Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One final-check and exposure-history record.

Compute metrics,judge workflow actions and write a bounded version-specific report.

Keep one habit: W1–W3/W5 → selected fresh counts → project freeze record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Train a Regression Model

> **My remembered idea, example and correction**

Mean, Median, and Baselines

> **My remembered idea, example and correction**

Precision and Recall

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main frozenV1recordF01–F20:TN12,FP2,FN3,TP3,accuracy15/20=75%,precision3/5=60%,recall3/6=50%;previously fitted clear baseline14/20=70%. Main mistakenIDsF13,F14falsealarms,F15–F17misses. Picture insteadTP3,FN1,FP2,TN6:9/12,3/5,3/4. Test-informed threshold changes make this development evidence for the revised version;retain the historical V1report and state no independent V2estimate if no fresh cases exist.

  Worked public fixtures are not hidden final tests for new learner models.

- Reference for this exercise: open this week's student lesson, "Hold Out Test Data", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate the worked accuracy, precision and recall with denominators.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main frozenV1recordF01–F20:TN12,FP2,FN3,TP3,accuracy15/20=75%,precision3/5=60%,recall3/6=50%;previously fitted clear baseline14/20=70%. Main mistakenIDsF13,F14falsealarms,F15–F17misses. Picture insteadTP3,FN1,FP2,TN6:9/12,3/5,3/4. Test-informed threshold changes make this development evidence for the revised version;retain the historical V1report and state no independent V2estimate if no fresh cases exist.

  Worked public fixtures are not hidden final tests for new learner models.

- Why does 75 percent versus 70 percent not settle deployment?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main frozenV1recordF01–F20:TN12,FP2,FN3,TP3,accuracy15/20=75%,precision3/5=60%,recall3/6=50%;previously fitted clear baseline14/20=70%. Main mistakenIDsF13,F14falsealarms,F15–F17misses. Picture insteadTP3,FN1,FP2,TN6:9/12,3/5,3/4. Test-informed threshold changes make this development evidence for the revised version;retain the historical V1report and state no independent V2estimate if no fresh cases exist.

  Worked public fixtures are not hidden final tests for new learner models.

- The team changes a threshold after reading test errors. How should it label later results?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main frozenV1recordF01–F20:TN12,FP2,FN3,TP3,accuracy15/20=75%,precision3/5=60%,recall3/6=50%;previously fitted clear baseline14/20=70%. Main mistakenIDsF13,F14falsealarms,F15–F17misses. Picture insteadTP3,FN1,FP2,TN6:9/12,3/5,3/4. Test-informed threshold changes make this development evidence for the revised version;retain the historical V1report and state no independent V2estimate if no fresh cases exist.

  Worked public fixtures are not hidden final tests for new learner models.

- One planned case returns no **prediction**. What must the report retain?

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the selected TN7,FP1,FN2,TP2for12new fictional cases,with blocked positive and clear/blocked reference-row/prediction-column axes. These differ from both the20-case main fixture and the pictured12-leaf counts. State the specific frozen-version scope and missing source-condition detail rather than claiming dependable performance in every setting.

- A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Compare a baseline for Final testing and honest **claim**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A final-test folder has twenty leaves; its labels are opened weekly to choose models.
- Is it still untouched?
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 23 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Open the final hold-out only after recording the frozen model choice.
- Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.
- What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/23

Frozen final-test results

	predicted spotted	predicted healthy
actual spotted	3	1
actual healthy	2	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/23

Hold Out Test Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record freeze

Record model version and choices before reading these final references.

2. Calculate metrics

Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.

3. Bound claim

Report these twelve held-out cases and their collection conditions; preserve all three error records.

CHANGE ONE THING

The team changes a threshold after inspecting the false negatives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Open the final hold-out only after recording the frozen model choice.
- Which labelled step matches this detail? Report these twelve held-out cases and their collection conditions; preserve all three error records.
- 1. Record **freeze**
- 2. Bound **claim**
- 3. Calculate metrics
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Bound **claim** Report these twelve held-out cases and their collection conditions; preserve all three error records. Read the picture in order.

Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The team changes a threshold after inspecting the false negatives. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This set is now development **evidence** for the revised model.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: This set is now development **evidence** for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Repair and keep the record

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

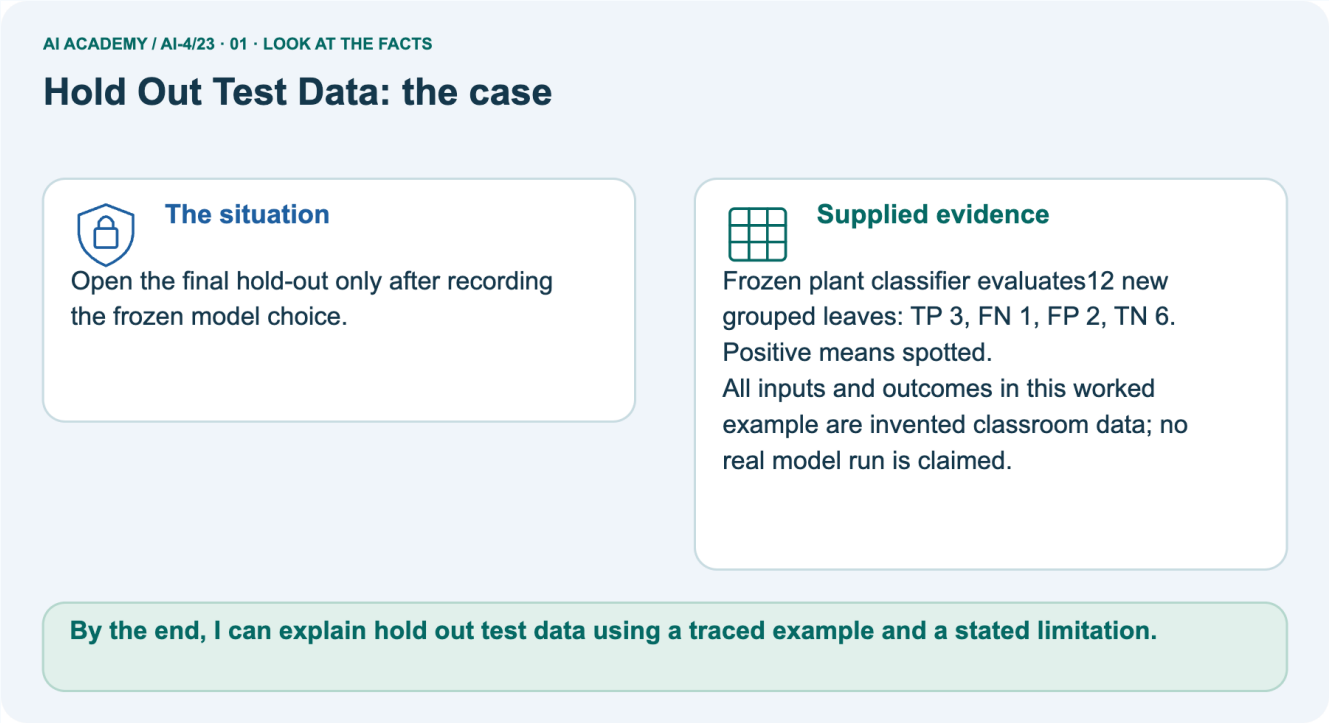
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Hold Out Test Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

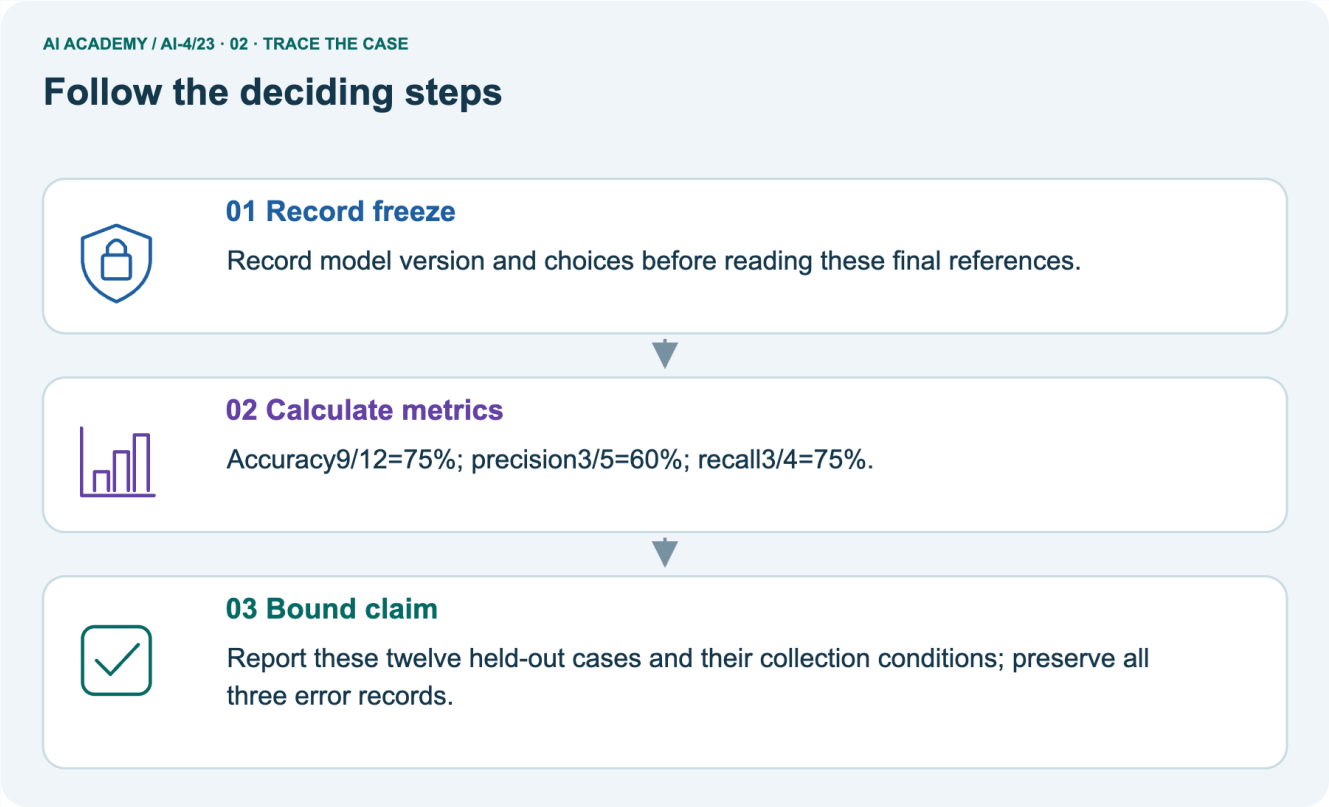
Open the final hold-out only after recording the frozen model choice. Supplied facts: Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



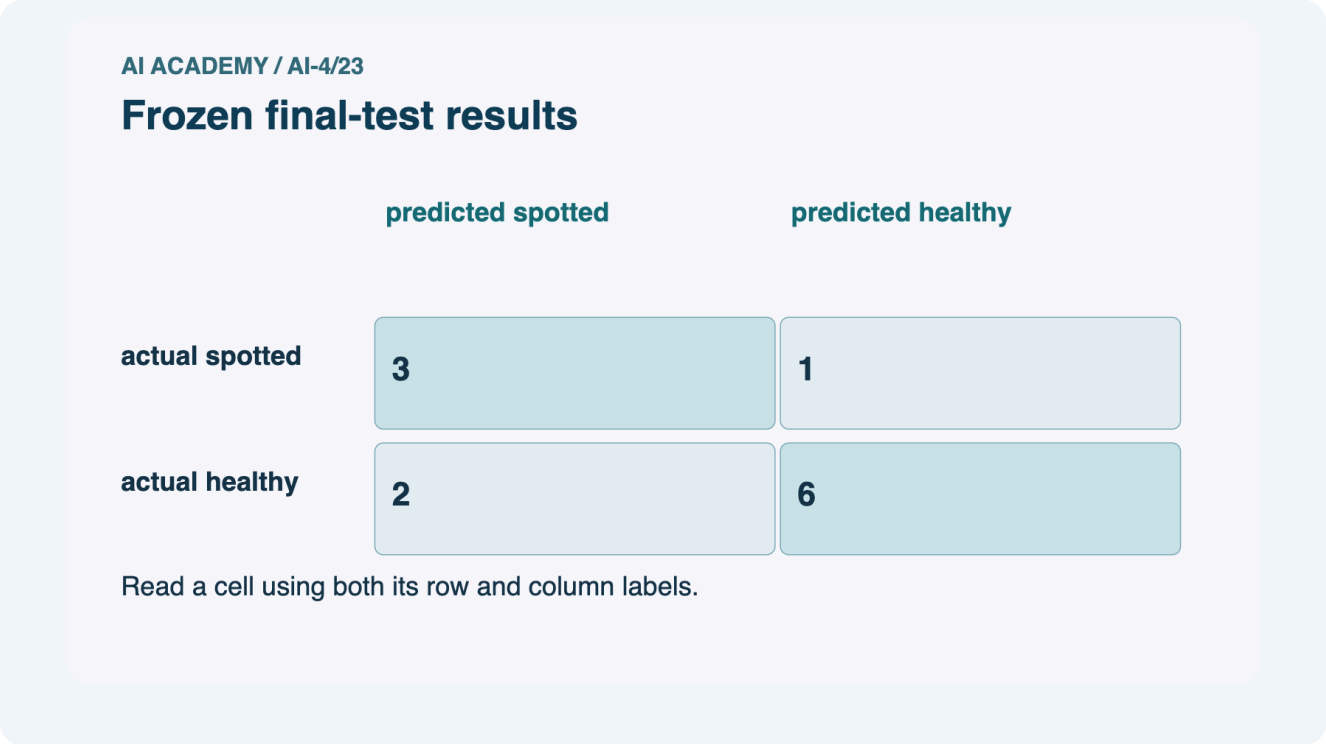
Follow the labelled picture with the matching explanation. Open the vector diagram

Record **freeze**: Record model version and choices before reading these final references. → Calculate metrics: Accuracy9/12=75%; precision3/5=**60%**; recall3/4=75%. → Bound **claim**: Report these twelve held-out cases and their collection conditions; preserve all three error records.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Frozen final-test results



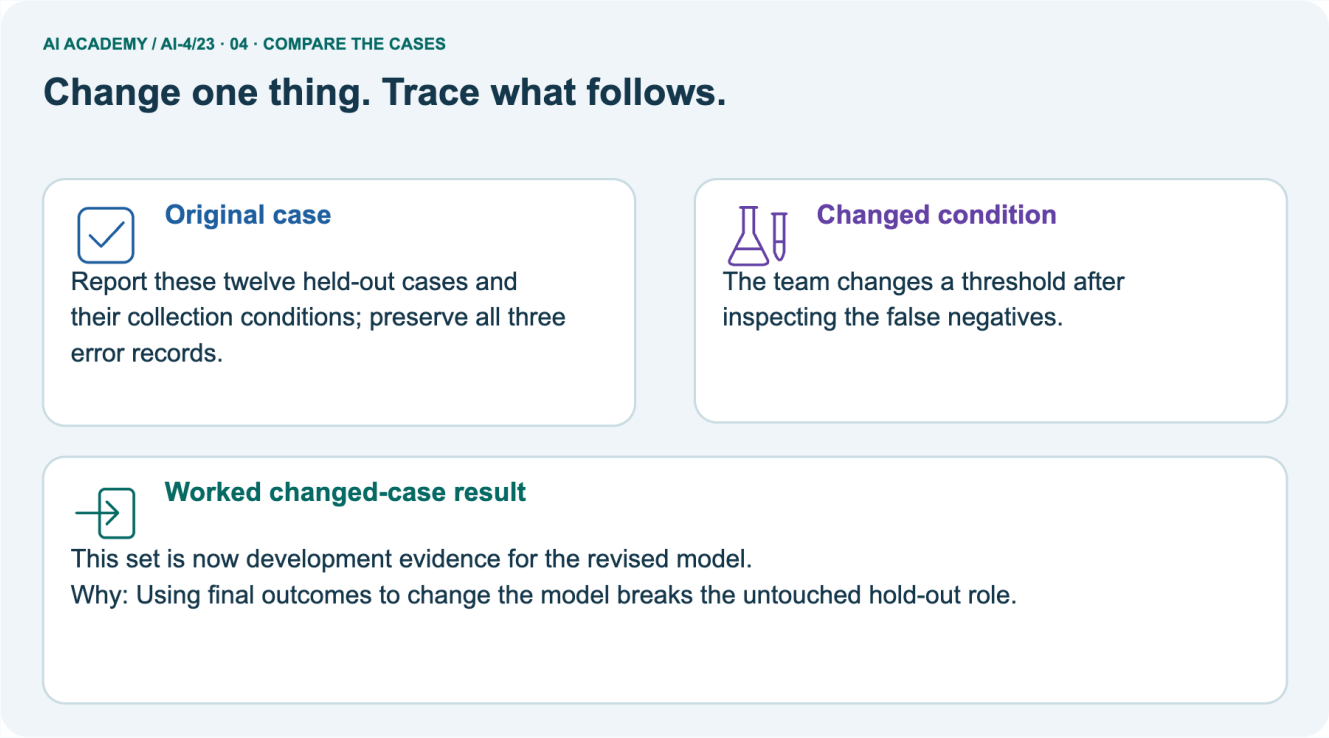
Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen final-test results. actual spotted: predicted spotted = 3, predicted healthy = 1. actual healthy: predicted spotted = 2, predicted healthy = 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

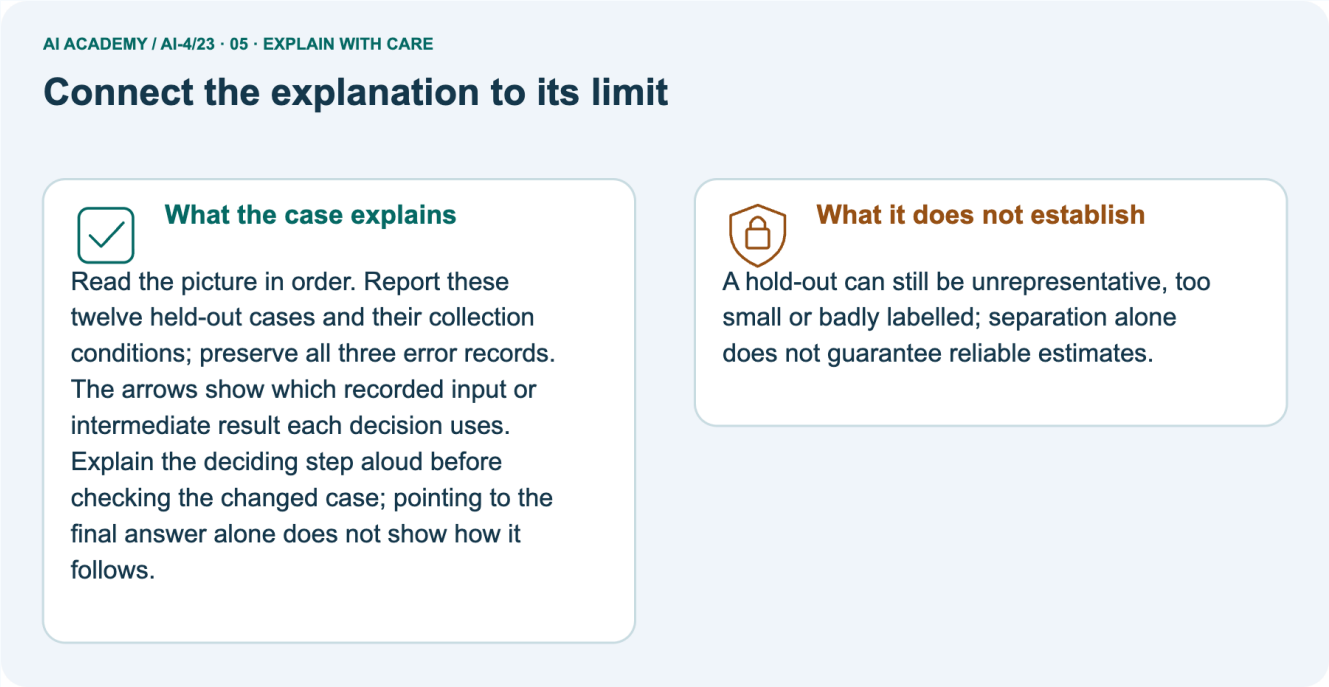
Original case: Report these twelve held-out cases and their collection conditions; preserve all three error records. Changed condition: The team changes a threshold after inspecting the false negatives.

Expected result: This set is now development **evidence** for the revised model. Reason: Using final outcomes to change the model breaks the untouched hold-out role.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

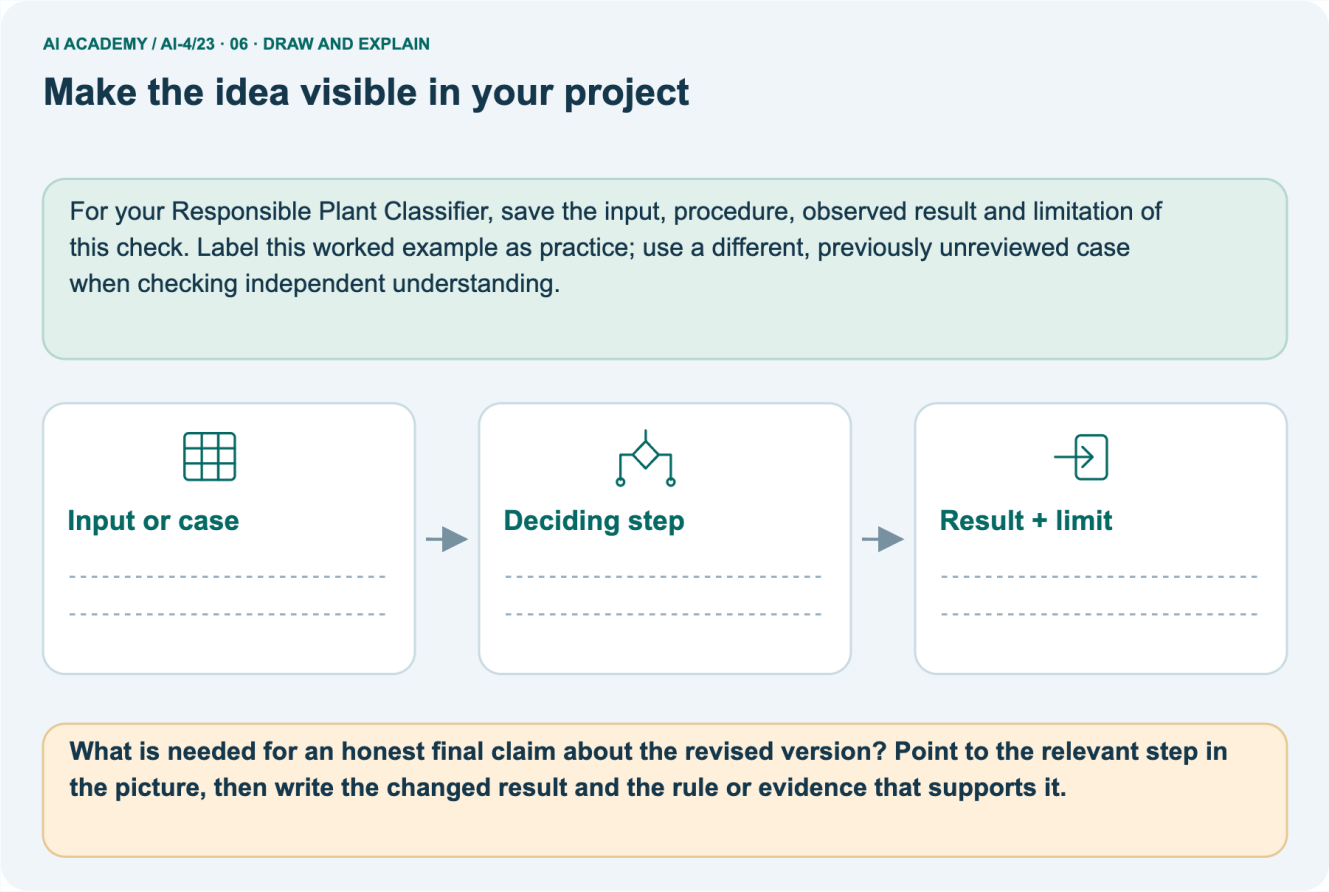
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Explain the case, then point to a limit on the **claim**.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Hold Out Test Data: the case

02. Follow the deciding steps

03. Frozen final-test results

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.